

Contribution Report

Deep Learning Final Project

Real-Time Financial Fraud Detection with Learned Hybrid Gating and Uncertainty-Aware Triage

AI Academy, National Artificial Intelligence Center, Azerbaijan

11 September 2026

This report summarizes the agreed division of work and principal contributions of the four project members. The project was developed collaboratively, with each member owning a major technical component while contributing to integration, testing, experimentation, documentation, and final project delivery.

Member Name	Code	Report Sections	Experiments	Slides	Other
Əskərov Nadir	Data pipeline, temporal splitting, pre-processing, historical sequence extraction, dataset/DataLoader components, classical baselines, evaluation metrics and cost analysis.	Dataset protocol, chronological splitting, sequence protocol, baseline methodology and results.	Baseline benchmarking and historical sequence-window ablation.	Dataset characteristics, sequence representation, and baseline results.	Prepared the leakage-aware data foundation and processed inputs used by downstream models.
Fətəliyev Akşin	FT-CAT Transformer, feature tokenization, self-attention and temporal cross-attention, focal-loss components, and supervised transformer training.	FT-CAT architecture, attention methodology, focal-loss design, and supervised deep-learning methodology/results.	FT-CAT architecture ablation and focal-loss sensitivity analysis assigned as the supervised-model experiments.	Primary ownership and coordination of the final 15–20 minute presentation deck, including architecture and model-result slides.	Produced the supervised deep-learning branch and trained FT-CAT model checkpoint.
Nəzərov Səbuhi	Denoising autoencoder, autoencoder training, centralized utilities, serving schemas/API, model serialization, Docker/CI infrastructure, and reproduction tooling.	DAE methodology, serving architecture, deployment notes, and latency evaluation.	Autoencoder anomaly evaluation and latency/throughput evaluation.	DAE, serving, deployment, and system-integration material.	Developed the anomaly-detection and serving/deployment foundation, including FastAPI, serialization, containerization, CI/CD, and reproduction tooling.
Hüseynov Habil	Learned hybrid gating, conformal prediction and triage, SHAP explainability, Streamlit analyst dashboard, integration and robustness testing, and supporting evaluation/serialization work.	Introduction, related work, learned gating, conformal prediction, XAI analysis, results interpretation, discussion/limitations, conclusion, and final IEEE paper assembly.	Learned-vs-fixed gating evaluation, multi-seed gate robustness, conformal coverage/workload evaluation, SHAP consistency analysis, DAE latent-space t-SNE, and CPU inference benchmarking/integration.	Conformal triage, learned gating, XAI, dashboard, and final result/deployment material.	Compiled the final IEEE report; implemented and validated the analyst dashboard; performed final Member-D integration, regression testing, artifact validation, and contribution-report assembly.

Individual Contribution Summary

Əskərov Nadir — Data & Sequence Lead.

Responsible for the data and sequence-processing foundation of the system, including leakage-aware chronological splitting, pre-processing, historical transaction sequence construction, classical machine-learning baselines, and core evaluation definitions.

Fətəliyev Akşin — Supervised Transformer Lead.

Responsible for the supervised FT-CAT branch, including feature tokenization, self-attention, temporal cross-attention, focal-loss formulation, supervised training, and coordination of the final presentation deck.

Nəzərov Səbuhi — Self-Supervised Anomaly & Serving Lead.

Responsible for the denoising autoencoder and the principal engineering and deployment infrastructure, including centralized utilities, FastAPI serving, model serialization, Docker/CI integration, and reproducibility tooling.

Hüseynov Habil — Learned Gating, Conformal & UI Lead.

Responsible for learned hybrid fusion, uncertainty-aware conformal triage, DAE-focused SHAP explainability, the Streamlit analyst interface, latent-space visualization, gating and conformal experiments, robustness evaluation, final IEEE report compilation, and contribution-report assembly.

Team Agreement

The members acknowledge that the contribution summary above reflects the agreed division of responsibilities and project work. The system was integrated and evaluated collaboratively, and all members contributed to the completion of the final project. The names below identify the project members associated with this contribution report.

MEMBER ACKNOWLEDGEMENTS

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Əskərov Nadir

DATA & SEQUENCE LEAD

MEMBER ACKNOWLEDGEMENT

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Hüseynov Habil

LEARNED GATING, CONFORMAL & UI LEAD

MEMBER ACKNOWLEDGEMENT

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Fətəliyev Akşin

SUPERVISED TRANSFORMER LEAD

MEMBER ACKNOWLEDGEMENT

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Nəzərov Səbuhi

SELF-SUPERVISED ANOMALY & SERVING LEAD

MEMBER ACKNOWLEDGEMENT

The member names and initials above serve as typed acknowledgements associated with this contribution report.